PULLBACK OF A PERFECT MODULE OF TOR DIMENSION AT MOST ONE ALONG A BLOWUP IS ALREADY DERIVED AND STAYS PERFECT OF TOR DIMENSION AT MOST ONE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; no macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0ESS (sources/stacks-flat/pinned-original-source.tex line 13224).

Allowed standard prerequisites (editorial).

- Perfect modules of tor dimension at most one and their two-term locally free presentations. Same chapter, flat.tex, the material on perfect modules and tor dimension preceding this lemma (the statement uses tor dimension <= 1 and perfectness without recalling them). Supplies that a perfect module of tor dimension <= 1 admits locally a two-term presentation by finite free modules, which is the presentation used at proof lines 13234-13237.
- The affine blowup algebra A[I/a] and its relation to the localization A_a. Divisors, Section on blowing up (Stacks tags 0804-0806 region), specifically the definition of the affine blowup algebra as a subring of A_a. Supplies that A[I/a] is a subring of A_a so that the localization argument at proof lines 13239-13242 applies; the blowup itself is only the ambient construction, not a counted step.

Lemma 0.1. *Let X be a scheme. Let $\mathcal{F}$ be a perfect $\mathcal{O}_X$-module of tor dimension ≤ 1. For any blowup $b : X' \to X$ we have $Lb^*\mathcal{F} = b^*\mathcal{F}$ and $b^*\mathcal{F}$ is a perfect $\mathcal{O}_X$-module of tor dimension ≤ 1.*

Proof. We may assume $X = \mathrm{Spec}(A)$ is affine and we may assume the A-module M corresponding to $\mathcal{F}$ has a presentation

$$0 \to A^{\oplus m} \to A^{\oplus n} \to M \to 0$$

Suppose $I \subset A$ is an ideal and $a \in I$. Recall that the affine blowup algebra $A[\frac{I}{a}]$ is a subring of A_a. Since localization is exact we see that $A_a^{\oplus m} \to A_a^{\oplus n}$ is injective. Hence $A[\frac{I}{a}]^{\oplus m} \to A[\frac{I}{a}]^{\oplus n}$ is injective too. This proves the lemma. □

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